

Sai Manikanta Karnati

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years building scalable backend systems and cloud-native services across Python, Java, Kotlin, and C. Skilled in REST APIs, MySQL, AWS, Kubernetes, and Machine Learning integrations. Delivered event-driven platforms at 99.99% reliability and cut operations effort 40% through automation, strong communication, and clean, maintainable code.

SKILLS

Python - Kotlin - C - Java - REST - MySQL - AWS - Kubernetes - Machine Learning - Spring Boot - Microservices - Node.js - TypeScript - Angular - GraphQL - Docker - Terraform - Apache Kafka - CI/CD - DynamoDB - Redis - GCP - Agile - Operations - Technical Writing

EXPERIENCE

Software Engineer - Vanguard

08/2025 - present

- Architected scalable backend services in Python, Java Spring Boot, and Kotlin microservices, enabling independent deployments across the engineering code base.
- Built Python data processing pipelines and automation scripts that reduced manual operations effort by ~40% and improved processing efficiency.
- Designed Machine Learning-powered backend services integrating Python LLM APIs for intelligent text classification, training, and automated workflows.
- Optimized complex MySQL and NoSQL queries, stored procedures, and data models, improving reporting and transaction performance by 35%.
- Engineered high-throughput event-driven systems using DynamoDB Streams, Python/Node.js Lambda, and SNS/SQS, ensuring 99.99% reliability.
- Developed REST and GraphQL APIs consumed by Angular frontends, with C-based components for performance-critical paths.
- Deployed cloud-native applications on AWS (Lambda, ECS, EC2, S3, RDS) using Kubernetes, cutting production issue resolution time 40% via Splunk observability.

Associate Software Engineer - Xebia

01/2022 - 12/2023

- Developed backend applications in Python, Java Spring Boot, Kotlin, and Node.js, delivering scalable REST APIs and microservices.
- Built Python automation scripts for data transformation, validation, and backend operations orchestration, reducing manual effort ~30%.
- Optimized MySQL queries, joins, views, and stored procedures, improving API response times by 35%.
- Implemented event-driven architecture with Apache Kafka, reducing inter-service dependency failures by 35%.
- Containerized applications using Docker and Kubernetes, improving CI/CD deployment efficiency ~25% and provisioning infrastructure with Terraform on AWS and GCP.
- Created and documented REST APIs using Swagger/OpenAPI, applying clear technical writing and communication to enforce versioning and backward compatibility.
- Integrated authentication using OAuth2, JWT, and Spring Security, and applied Redis caching to boost throughput under heavy load.

Engineer Trainee - Xebia

01/2020 - 12/2021

- Completed engineering training program, developing enterprise applications using Angular and Python/Java backend services.
- Built and optimized MySQL and MongoDB queries, stored procedures, and relational schemas, improving query execution time by 20%.
- Developed REST APIs and documented service contracts using Swagger/OpenAPI, strengthening team communication.

- Implemented API security using OAuth2 and Spring Security, plus rate limiting and gateway protections for backend operations.
- Increased unit test coverage to 95% using JUnit and Python-based test automation, reinforcing a reliable code base.

EDUCATION

Master of Science in Computer Science - University Of North Carolina At Charlotte

CERTIFICATIONS

- AWS Certified Developer Associate
- MongoDB Certified Developer
- Infosys Certified Software Programmer
- NPTEL Certified Java Developer
- AWS Academy Graduate - AWS Academy Data Engineering